Ryan J. Chen

Aspiring Code Wizard🧙‍♂️

I am a Junior @ Duke that wants to improve how humans live now and in the future. I'm interested in Computer Vision, Signal Processing, X-tended Reality, Computer Graphics, and the medical applications of technology and software. I am always open to learning more about the world around us and how we process it. Feel free to reach out to me for questions, or just to talk!

👈 Check out my website to find out more about me, and some of the projects I'm working on.

Contact

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Website
https://rjaychen.github.io

Education

Aug. 2021- May 2025

BS in ECE and CS
+ minor in japanese

Duke University

Expertise

- Software Engineering
- XR App Development
- Data Analysis
- Machine Learning
- Software Quality
- Language Learning
- Embedded Software

Language

English - Native

Chinese - Native

Japanese - Intermediate

German - Beginner

Experience

Fall 2023 - Spring 2024
Gorlatova Lab I Durham, NC, USA

AR Application Developer 🎮😎

- Design of AR application in Unity using C++/C# to program therapeutic responses for patients with misophonia
- Worked with Microsoft Hololens to provide catered mixed reality experience for patients and subjects
- Implement edge-computing architecture for detection, matching, and identification of retinal landmarks
- Optimized feature detection using Computer Vision algorithms for robust hologram registration and AR guidance

Summer 2023 - Winter 2023
Align Technology I Morrisville, NC, USA

Software Quality Engineering Intern ⚙️🔍

- Development of DevOps app for wide-scale optimization of the software development pipeline (Atlassian APIs)
- Built document automation software for streamlining of QA/QMS processes using OCR technologies
- Standardized document formatting using automation to expedite future QA/QMS worktimes
- Used Test-Driven Development ensure quality of delivered application through integration and unit testing

Fall 2022 - Spring 2023
Big Ideas Lab I Durham, NC, USA

Undergraduate Researcher 🧠🔬

- Analysis of EEG signals from OpenBCI device during AR stimuli as part of pilot study on behavior therapy
- Programmed statistical analysis algorithms in python and MATLAB to decode emotions of subjects
- Leveraged python libraries and MATLAB plugins for advanced processing, analysis, and visualization of signals
- Conference paper @ IEEE VR 2023: XR Health Studies

Stuff I Do

Combat Robotics

Alpha Phi Omega (Service)

Taiwanese Students Association

Japanese Culture Club

Tools and Skills

Python
C++/C#
MATLAB
Java
Restful APIs, Node.js, MongoDB
Atlassian Suite & APIS